

GEC-CCP loss design

$$\ell_t = \text{Huber} + 0.25 L_{\text{boundary}} + 0.10 L_{\nabla} + 0.05 L_{\text{multi}}$$

Huber

robust point fit

Boundary × 0.25

protect sheath edges

Gradient × 0.10

preserve spatial shape

Multi-scale × 0.05

retain broad structure

$$L_{\text{CCP}} = \frac{1}{3} \left[\frac{\ell_{n_e} + \ell_{n_i}}{2} + \ell_{T_e} + \ell_{\phi} \right]$$

equal weight for density, temperature, and electric potential